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Roll No

AU-5001-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Machine Component Design

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) PSG design data book and/ or Mahadevan / Reddy's Mechanical design data book can be provided/ permitted in examination hall.

1. a) What are the steps involved in design of a machine element? 7
b) What is factor of safety? Why is it necessary to use factor of safety? How will you find out allowable stress for ductile parts and brittle parts using factor of safety? 7
2. a) How the shaft is designed when it is subjected to bending moment only? 4
b) A shaft supported at the ends in ball bearings carries a straight tooth spur gear at its mid span and is to transmit 7.5 kW at 300 r.p.m. The pitch circle diameter of the gear is 150mm. The distances between the center line of bearings and gear are 100mm each. If the shaft is made of steel and the allowable shear stress is 45 MPa, determine the diameter of the shaft. Show in a sketch how the gear will be mounted on the shaft; also indicate the ends where the bearings will be mounted? The pressure angle of the gear may be taken 20°. 10

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3. a) How are the keys classified? Draw neat sketches of different types of keys and state their application. 7
b) It is required to design a square key for fixing a gear on a shaft, of 25 mm diameter. The shaft is transmitting 15 kW power at 720 r.p.m. to the gear. The key is made of steel 50C4 ($S_{yt} = 460 \text{ N/mm}^2$) and the factor of safety is 3. For key material, the yield strength in compression can be assumed to be equal to the yield strength in tension. Determine the dimensions of the key. 7
4. a) What is overhauling of power screw? What is the condition for overhauling? 4
b) A rigid coupling is used to transmit 50 kW power at 300 rpm. There are six bolts. The outer diameter of the flanges is 200 mm, while the recess diameter is 150 mm. The coefficient of friction between the flanges is 0.15. The bolts are made of steel 45C8 ($S_{yt} = 380 \text{ N/mm}^2$) and the factor of safety is 3. Determine the diameter of the bolts. Assume that the bolts are fitted in large clearance holes. 10
5. a) Name the different types of clutches. Describe with the help of neat sketch the working principle of single plate clutch. 7
b) A plate clutch consists of one pair of contacting surfaces. The inner and outer diameters of the friction disk are 100 and 200 mm respectively. The coefficient of friction is 0.2 and the permissible intensity of pressure is 1 N/mm^2 . Assuming uniform-wear theory; calculate the power transmitting capacity of the clutch at 750 rpm. 7
6. a) What is the function of brake? State different types of brakes and give at least one practical application of each. 7

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- b) A rope drum of an elevator having 650 mm diameter is fitted with a brake drum of 1 m diameter. The brake drum is provided with four cast iron brake shoes each subtending an angle of 45° . The mass of the elevator when loaded is 2000 kg and moves with a speed of 2.5 m/s. The brake has a sufficient capacity to stop the elevator in 2.75 meters. Assuming the coefficient of friction between the brake drum and shoes as 0.2, find: (i) width of the shoe, if the allowable pressure on the brake shoe is limited to 0.3 N/mm^2 and (ii) heat generated in stopping the elevator. 7
7. a) Explain about the Lewis equation to determine the beam strength of gear teeth. 7
- b) A reciprocating compressor is to be connected to an electric motor with the help of spur gears. The distance between the shafts is to be 500 mm. The speed of the electric motor is 900 r.p.m. and the speed of the compressor shaft is desired to be 200 r.p.m. The torque, to be transmitted is 5000 N-m. Taking starting torque as 25% more than the normal torque, determine: (1) Module and face width of the gears using 20 degrees stub teeth, and (2) Number of teeth and pitch circle diameter of each gear. Assume suitable values of velocity factor and Lewis factor. 7
8. Write short notes on the followings: 14
- a) Working principle of flange coupling
 - b) Self-energizing block brake and self-locking block brake
 - c) Virtual number of teeth on a helical gear
